

Forecasting Realized Volatility of Indian Equity Indices Using Classical Time-Series Models: A Walk-Forward Comparative Study

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Abstract

We forecast the realized volatility (RV) of three Indian equity indices - NIFTY 50, SENSEX, and BANK NIFTY - using four classical time-series methods: the naïve (random-walk) benchmark, Simple Exponential Smoothing (SES), Holt's linear trend, and ARIMA. RV is constructed as the rolling standard deviation of daily log returns over four window lengths (5, 10, 20, 30 days), giving twelve index-window combinations. Models are evaluated by expanding-origin, one-step-ahead *walk-forward* validation and compared on RMSE, MAE, MASE, and sMAPE, with Diebold–Mariano (DM) tests, moving-block bootstrap confidence intervals, and full residual diagnostics. Our central finding is that *the ranking depends on the loss function*: under MASE the naïve random walk is not beaten (mean MASE 0.827 for naïve/SES versus 0.851 for ARIMA and 0.880 for Holt), whereas under squared-error loss ARIMA attains the lowest mean RMSE and is more accurate than naïve in 9 of 12 cells, significantly so (DM, $p < 0.05$) in 3. Residuals remain heteroskedastic and heavy-tailed in every case, and the stationarity tests conflict (ADF rejects a unit root while KPSS rejects stationarity), consistent with the strong, partly artificial persistence induced by overlapping windows. The results reproduce, on Indian data, the well-known difficulty of beating a simple benchmark in short-horizon volatility forecasting and motivate variance-based models (GARCH, HAR-RV) as the next step.

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1 Introduction

Forecasting the direction of asset prices is notoriously hard: efficient-market reasoning implies that the conditional mean of returns is close to unpredictable, so a model such as “ARIMA on tomorrow’s closing price” rarely improves on a random walk. Volatility is different. It is persistent, it clusters in time, and - crucially - it is the quantity that actually governs risk. This makes volatility both more forecastable and more useful than price direction, which is why we forecast *realized volatility* rather than price.

This study compares four classical time-series models on the realized volatility of three Indian equity indices. Our contributions are: (i) a controlled comparison across three indices and four volatility-window lengths under a rigorous walk-forward protocol; (ii) evaluation under four accuracy metrics together with formal Diebold–Mariano tests and bootstrap uncertainty; and (iii) an honest assessment of where classical models help and where they do not. We deliberately foreground a negative result - the naïve benchmark is hard to beat - because it is both statistically defensible and practically informative.

2 Financial Theory

2.1 Why log returns

Given a price P_t , the daily log return is

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right). \quad (1)$$

Log returns are time-additive (multi-day returns are sums of daily returns), approximately symmetric in gains and losses, and, if prices are approximately log-normal, approximately Gaussian - properties that make them the natural input to the models and diagnostics used here.

2.2 Why realized volatility

True volatility is a latent quantity: it is never observed directly. Realized volatility is a model-free *estimate* of it. Here we use the rolling standard deviation of returns over a window of w trading days,

$$\text{RV}_t^{(w)} = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} (r_{t-i} - \bar{r})^2}. \quad (2)$$

Rolling windows expose *volatility clustering* - the empirical tendency of turbulent and calm periods to persist - which is precisely the structure that makes forecasting worthwhile.

2.3 Why forecast volatility rather than price

Because prices behave close to a random walk, the mean of returns carries little forecastable signal. Volatility, by contrast, is persistent and mean-reverting (the ARCH effect) and drives Value-at-Risk, option pricing, and position sizing. A volatility forecast is therefore both more tractable and more economically relevant than a price forecast.

3 Time-Series Theory

3.1 Stationarity: ADF and KPSS together

An ARMA representation assumes the series has a stable mean, variance, and autocovariance. We test this with two complementary tests whose null hypotheses are opposite: the Augmented Dickey–Fuller (ADF) test has H_0 of a unit root (non-stationary), so small p favours stationarity; the KPSS test has H_0 of stationarity, so large p favours it. Using both distinguishes a genuinely stationary series from a trend- or difference-stationary one, and flags the ambiguous near-unit-root case in which the two tests disagree.

3.2 ACF, PACF and differencing

The autocorrelation function (ACF) identifies the moving-average order q ; the partial autocorrelation function (PACF) identifies the autoregressive order p . When a series is non-stationary in level, differencing (the d in ARIMA) removes the unit root, with the caution that over-differencing inflates variance and introduces spurious negative autocorrelation.

3.3 Model identification and residual diagnostics

Following Box–Jenkins, candidate orders are read from the ACF/PACF and compared using information criteria. We select by the Bayesian Information Criterion (BIC), which penalises parameters more heavily than AIC and favours parsimony. A well-specified model leaves residuals indistinguishable from white noise, which we assess with the Ljung–Box test (no autocorrelation), the Jarque–Bera test (normality), and the residual ACF.

3.4 Comparing forecasts: Diebold–Mariano and MASE

To decide whether one model is *significantly* more accurate than another we use the Diebold–Mariano test on the one-step loss differential. As a scale-free accuracy measure we use the Mean Absolute Scaled Error, which scales the test-set MAE by the in-sample MAE of a one-step naïve forecast; $\text{MASE} < 1$ means the model beats that naïve benchmark.

4 Data and Methodology

4.1 Data

Daily closing prices for NIFTY 50, SENSEX, and BANK NIFTY were obtained from Yahoo Finance from January 2015 onward. Log returns and the four RV series ($w \in \{5, 10, 20, 30\}$) were constructed as in Section 2. Figure 1 shows the representative RV level series; the pronounced 2020 spike (the COVID-19 shock) and the surrounding clusters are clearly visible.

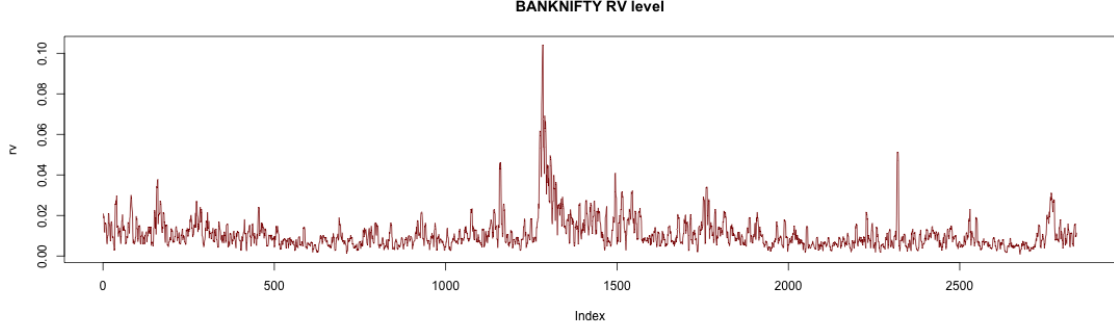


Figure 1: BANK NIFTY 5-day realized volatility (level). Volatility clustering and the large 2020 spike are evident; the series is strongly persistent.

4.2 Walk-forward evaluation

Each RV series is split chronologically, and forecasts are produced by expanding-origin, one-step-ahead walk-forward validation: model parameters (the ARIMA order and the smoothing constants) are estimated on the initial block and then rolled forward one observation at a time, so that every forecast uses only past information. This is the fair protocol for a persistent series; a single static multi-step forecast would collapse toward the mean and confound the comparison.

4.3 Models

Four methods are compared. The *naïve* forecast carries the last observed value forward. *SES* and *Holt* are exponential-smoothing models of level and of level-plus-trend, respectively. *ARIMA* orders are selected automatically per cell by BIC; the stationarity decision restricts the differencing search where it is conclusive.

4.4 Metrics

Models are scored by RMSE, MAE, MASE, and sMAPE. sMAPE is included because RV can be small, which makes the ordinary MAPE unstable. Uncertainty in the metrics is quantified by a moving-block bootstrap, and pairwise significance by the DM test.

5 Results

5.1 Stationarity

In every one of the twelve series the ADF test rejected a unit root ($p \approx 0.01$) while the KPSS test also rejected stationarity ($p < 0.05$). This is the classic conflicting-evidence case of a *near-unit-root* process: the series is highly persistent without being a pure random walk. The conflict is a direct consequence of the overlapping rolling windows, which mechanically inject autocorrelation. Because the decision was inconclusive, `auto.arima` was left to choose the differencing order, and it selected $d = 1$ in all twelve cells.

5.2 ACF, PACF and order selection

Figure 2 shows the slowly decaying ACF and the sharp PACF cut-off after lag 1 that are characteristic of a persistent, near-integrated series. The BIC-selected ARIMA orders (Table 1, Figure 3) are dominated by ARIMA(1, 1, 1) (five cells), with ARIMA(0, 1, 0) - the pure random walk - selected in the three 10-day cells and ARIMA(5, 1, 0) in the three 5-day cells. That BIC

chooses the random walk outright in a quarter of cases is itself evidence of how close these series are to non-forecastable.

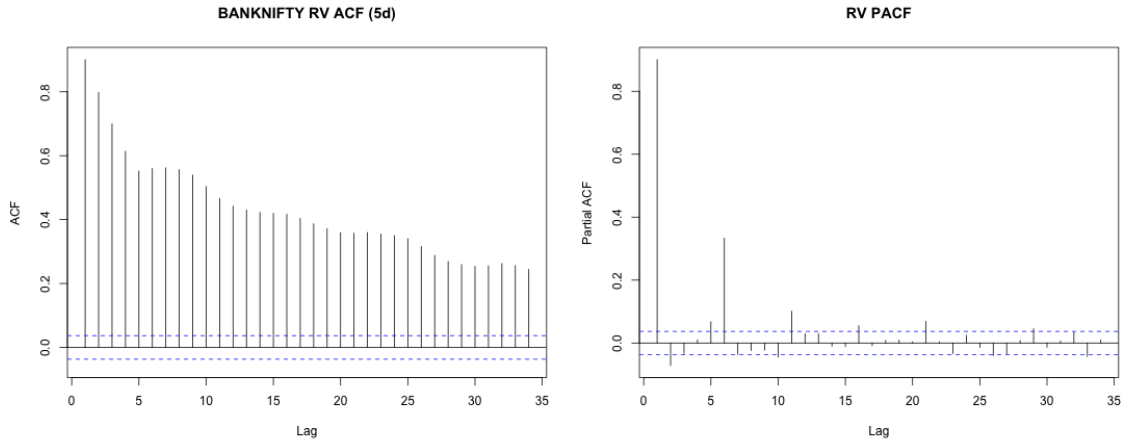


Figure 2: ACF (left) and PACF (right) of the representative RV series. The slow ACF decay and the dominant lag-1 PACF spike indicate strong persistence.

Index	$w = 5$	$w = 10$	$w = 20$	$w = 30$
NIFTY 50	(5,1,0)	(0,1,0)	(1,1,1)	(1,1,1)
SENSEX	(5,1,0)	(0,1,0)	(1,1,1)	(1,1,1)
BANK NIFTY	(5,1,0)	(0,1,0)	(1,1,1)	(2,1,2)

Table 1: BIC-selected ARIMA(p, d, q) orders. Frequency: $(1, 1, 1) \times 5$, $(5, 1, 0) \times 3$, $(0, 1, 0) \times 3$, $(2, 1, 2) \times 1$.

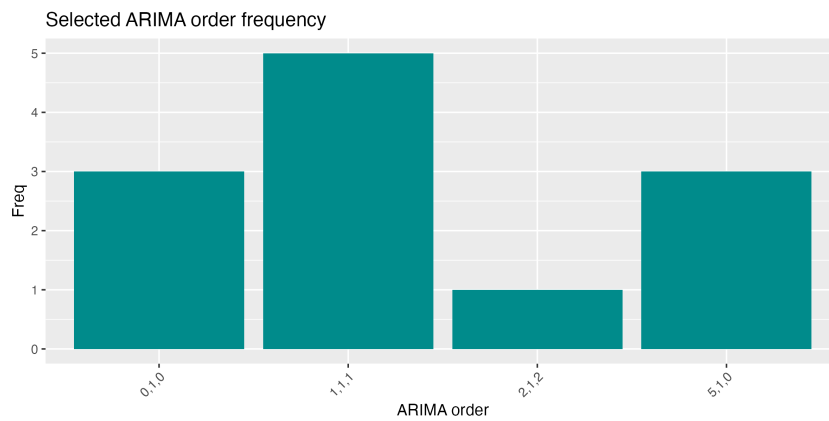


Figure 3: Frequency of BIC-selected ARIMA orders across the twelve cells.

5.3 Forecast accuracy

Table 2 reports the mean of each metric across all twelve cells, and this is where the central result appears: *the ranking depends on the loss function*. Under MASE and MAE the naïve random walk and SES are tied for best (mean MASE 0.827), ahead of ARIMA (0.851) and Holt (0.880); under RMSE, however, ARIMA attains the *lowest* mean error. The reason is visible in the data: ARIMA (especially the (5, 1, 0) specification at short windows) reduces the large errors around volatility spikes - which squared error penalises heavily - without improving the typical absolute error. Holt is consistently the weakest, its linear trend extrapolating poorly on a mean-reverting series.

Method	RMSE ($\times 10^{-3}$)	MAE ($\times 10^{-3}$)	MASE	sMAPE (%)
Naïve	1.262	0.690	0.827	8.78
SES	1.263	0.690	0.827	8.77
ARIMA	1.220	0.698	0.851	8.95
Holt	1.268	0.709	0.880	9.02

Table 2: Mean accuracy across all twelve index-window cells. Naïve/SES lead on MASE and MAE; ARIMA leads on RMSE. Lower is better throughout.

The full MASE grid (Table 3, Figure 4) shows the pattern is systematic rather than an averaging artefact: naïve/SES win in most cells, ARIMA is competitive at short windows, and Holt deteriorates at the 20-day window. The mean-MASE bar chart (Figure 5) confirms how narrowly separated the four methods are overall.

Index	Window	Naïve	SES	Holt	ARIMA
BANK NIFTY	5	0.760	0.761	0.761	0.769
BANK NIFTY	10	0.771	0.771	0.771	0.771
BANK NIFTY	20	0.791	0.791	0.886	0.826
BANK NIFTY	30	0.764	0.764	0.863	0.828
NIFTY 50	5	0.858	0.858	0.865	0.855
NIFTY 50	10	0.854	0.854	0.854	0.854
NIFTY 50	20	0.874	0.874	0.980	0.919
NIFTY 50	30	0.827	0.827	0.935	0.877
SENSEX	5	0.872	0.872	0.872	0.865
SENSEX	10	0.856	0.856	0.856	0.856
SENSEX	20	0.887	0.887	0.994	0.934
SENSEX	30	0.813	0.813	0.920	0.863

Table 3: MASE by index, window, and method. All values exceed 1 nowhere and cluster near 0.8–0.9; no method reliably dominates.

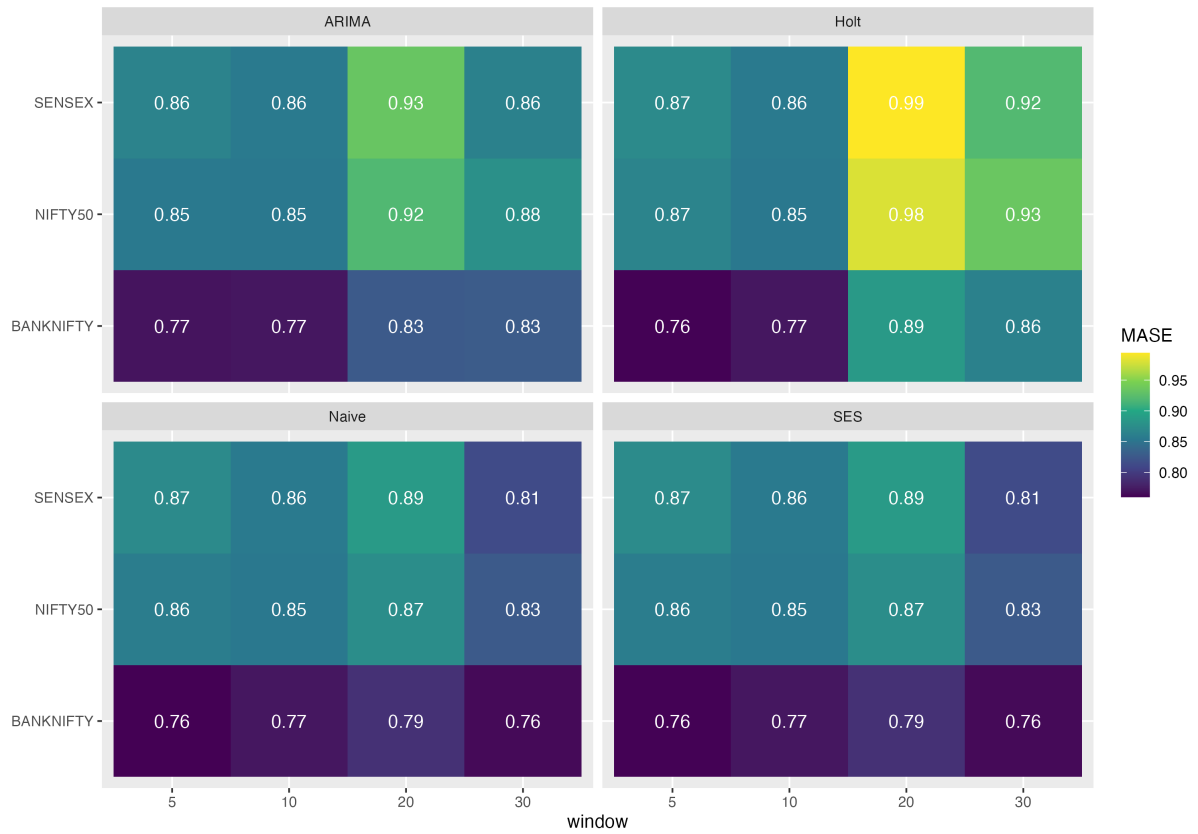


Figure 4: MASE by index (rows) and window (columns) for each method. Darker is lower (better). BANK NIFTY is the easiest series; Holt is worst at the 20-day window.

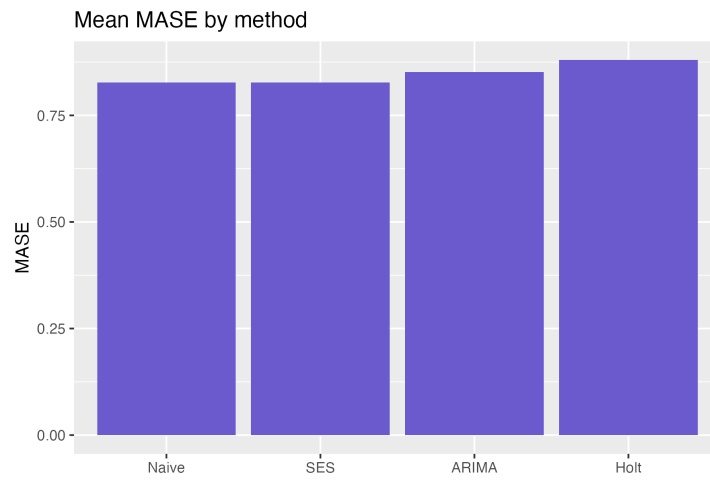


Figure 5: Mean MASE by method. The four methods are closely bunched, with naïve/SES marginally ahead.

5.4 Statistical significance (Diebold–Mariano)

On squared-error loss, ARIMA has a lower RMSE than naïve in 9 of 12 cells, but the DM test judges the improvement statistically significant ($p < 0.05$) in only 3 - the three short-window ARIMA(5, 1, 0) cells. Neither SES nor Holt significantly beats naïve in any cell. In other words, most of ARIMA’s apparent edge is within sampling noise, and where it is real it is confined to short windows where spike-smoothing helps.

5.5 Representative case: forecast, intervals, and diagnostics

The representative cell - selected automatically as the lowest-MASE cell among those where ARIMA significantly beats naïve - is BANK NIFTY at the 5-day window, with ARIMA(5, 1, 0). Figure 6 shows its one-step forecasts with 95% prediction intervals; empirical coverage is 97.4%, close to (and slightly conservative relative to) the nominal level. The forecast tracks the level well but, as expected for a one-step model, lags sharp jumps such as the mid-sample spike.

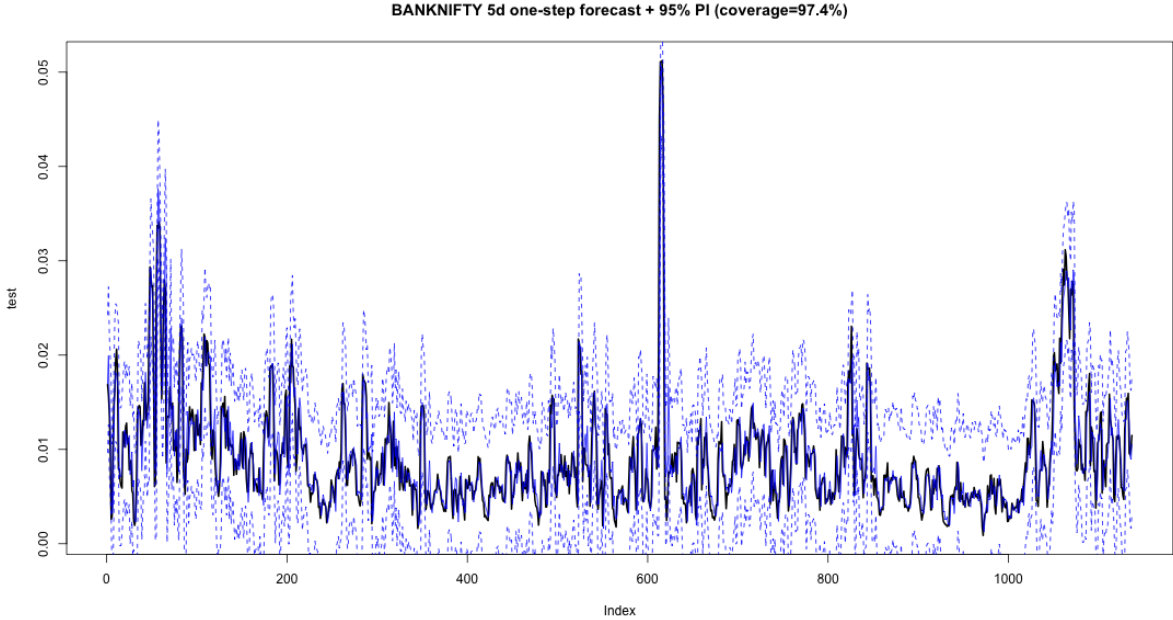


Figure 6: Representative one-step walk-forward forecast (solid) with 95% prediction intervals (dashed); empirical coverage 97.4%.

Residual diagnostics (Figure 7) tell the crucial cautionary story. The Ljung–Box test rejects white-noise residuals ($p < 0.001$) and the Jarque–Bera test rejects normality ($p < 0.001$): the residual histogram is sharply peaked with heavy tails, and the residual variance visibly expands around turbulent periods. ARIMA has captured the linear autocorrelation but not the conditional heteroskedasticity - exactly the structure a variance model is designed for. The per-method error histograms (Figure 8) show the same heavy-tailed shape for all four methods.

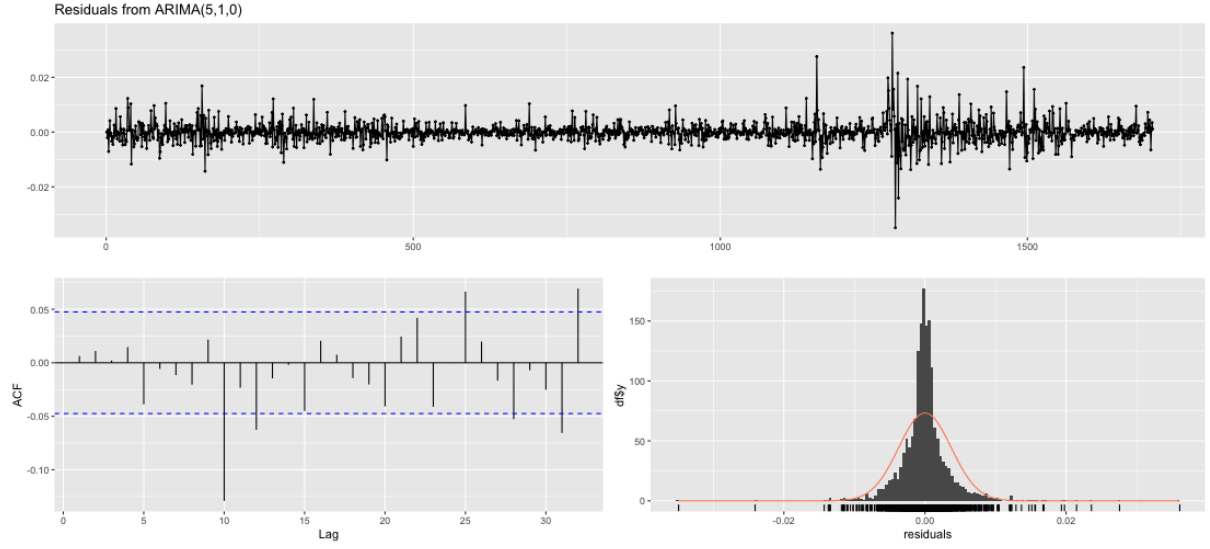


Figure 7: Residual diagnostics for the representative ARIMA(5,1,0): time plot, ACF, and histogram. The heavy tails and time-varying spread indicate leftover conditional heteroskedasticity.

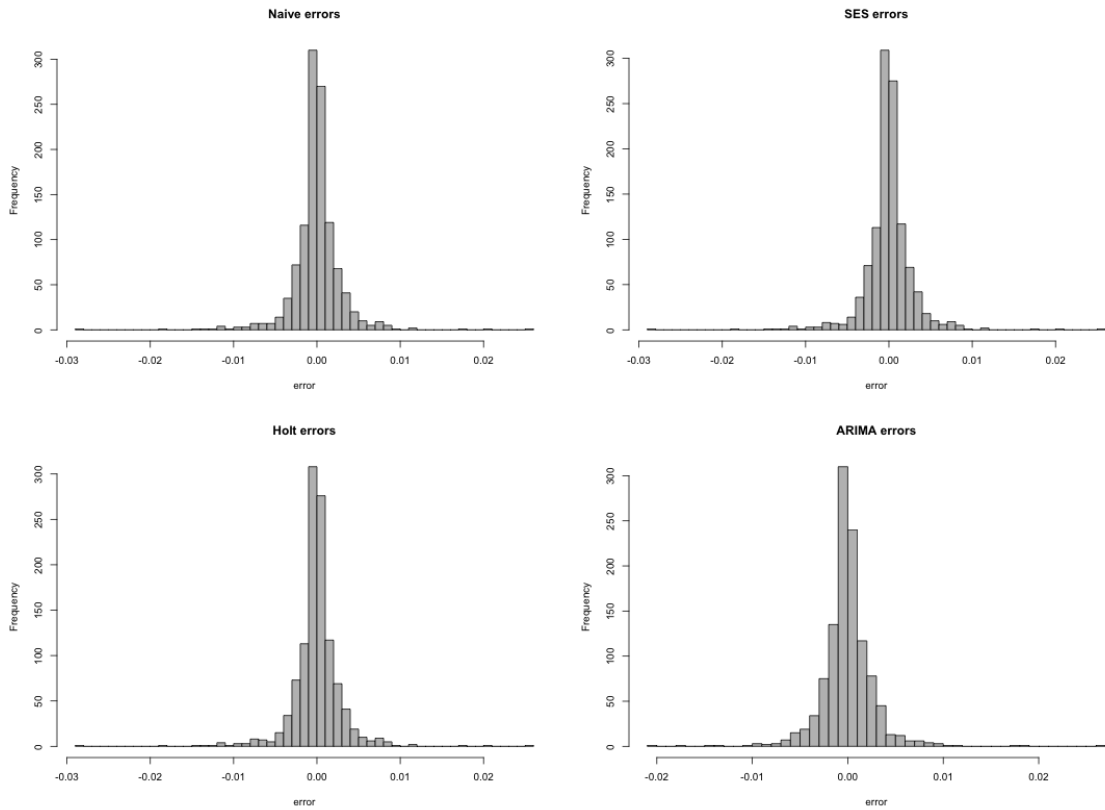


Figure 8: One-step forecast-error distributions for the four methods (representative cell). All are sharply peaked with heavy tails.

5.6 Uncertainty and robustness

Moving-block bootstrap confidence intervals for the representative cell (Table 4) overlap substantially between ARIMA and naïve, reinforcing that their difference is not practically meaningful there. The rolling-RMSE plot (Figure 9) shows all four methods moving almost in lockstep through time, with ARIMA’s small advantage concentrated in the high-volatility episodes. A sensitivity analysis over training fractions of 60%, 70%, and 80% (Figure 10) leaves the method ordering essentially unchanged. Finally, the window/accuracy relationship is weak and statistically insignificant (Spearman $\rho = 0.32$, $p = 0.304$): MASE *tended* to rise with window length, but the trend is not established. Across the twelve cells, AIC and BIC selected different orders in 9, yet mean MASE differed only modestly (0.851 for BIC versus 0.872 for AIC), so the criterion choice does not materially change conclusions.

Method	MASE 95% CI	Point MASE
ARIMA	[0.690, 0.860]	0.769
Naïve	[0.675, 0.847]	0.760

Table 4: Moving-block bootstrap 95% confidence intervals for MASE, representative cell. The intervals overlap heavily.

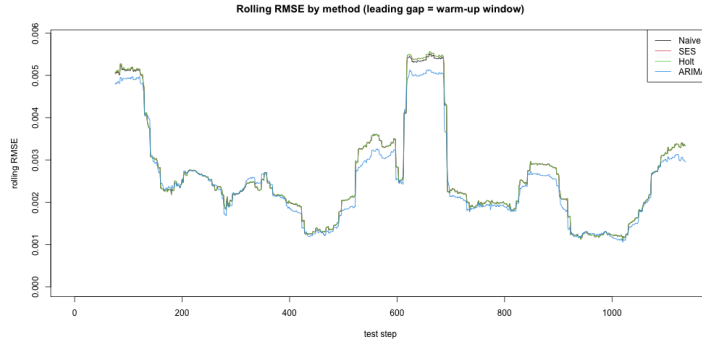


Figure 9: Rolling RMSE over the test period. The four methods track one another closely; ARIMA’s edge is concentrated in high-volatility episodes. The leading gap is the warm-up window.

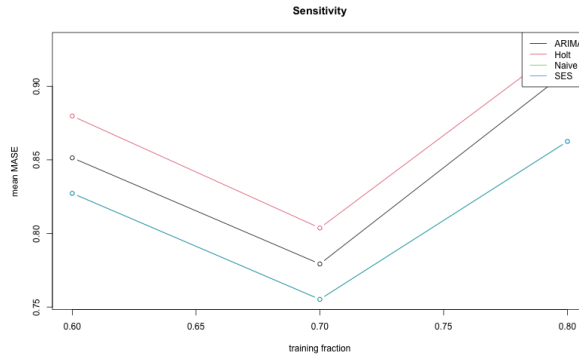


Figure 10: Sensitivity of mean MASE to the training fraction. The method ordering is stable across 60%, 70%, and 80%.

6 Discussion

The evidence converges on one message: for short-horizon realized-volatility forecasting on these indices, classical models do not deliver a robust improvement over the naïve random walk. Where ARIMA helps, it helps on squared-error loss by damping large errors around spikes, and only significantly at short windows. This is not a failure of the experiment but a substantive finding, and it echoes the volatility-forecasting literature: found that little beats a simple GARCH(1,1) for return volatility, and the near-random-walk behaviour of persistent volatility proxies is well documented.

Practically, the implication is that these forecasts are valuable for risk *measurement* - feeding Value-at-Risk and expected-shortfall estimates, option pricing (realized versus implied volatility), volatility-targeting position sizing, and margining - but they offer no directional edge over simply carrying yesterday's volatility forward. The strong, partly mechanical persistence of overlapping-window RV both makes the naïve benchmark hard to beat and inflates the apparent autocorrelation that a linear model exploits.

7 Limitations and Future Work

Several limitations frame these results. First, ARIMA models the conditional *mean*, whereas volatility clustering is a property of the conditional *variance*; ARIMA captures it only indirectly through the RV proxy, and the residual diagnostics confirm the cost in leftover heteroskedasticity and heavy tails. Second, the overlapping rolling-window construction injects artificial autocorrelation that inflates persistence and flatters the naïve benchmark. Third, ARIMA is linear and symmetric and cannot represent the leverage effect (volatility responding more to negative than to positive returns).

The natural next steps address the conditional variance directly: GARCH-family models (GARCH for clustering; EGARCH and GJR-GARCH for asymmetry) as in and ; the HAR-RV model of, which is purpose-built for realized volatility and its long memory; stochastic-volatility models that treat volatility as a latent process; and non-overlapping or higher-frequency RV estimators to remove the window artefact. Benchmarking the present ARIMA baseline against GARCH(1,1) and HAR-RV is the immediate priority.

8 Conclusion

Across three Indian indices and four volatility windows, evaluated by a rigorous walk-forward protocol, no classical model reliably beat the naïve random walk for one-step realized-volatility forecasting. The ranking is loss-function dependent: naïve and SES lead on MASE and MAE, while ARIMA edges ahead on RMSE by damping volatility spikes, significantly so only at short windows. Residuals remain heteroskedastic and heavy-tailed throughout. The study thus reproduces, on Indian data, a robust stylised fact of volatility forecasting and points clearly toward conditional-variance models as the productive next step.